

1- A solid rocket motor burns at the following rates at an initial temperature of 0°C in northern Canada.

$$r=2.5 \text{ mm/s at 15 atm}$$

$$r=7.0 \text{ mm/s at 60 atm}$$

The sensitivity of the propellant is:

$$\pi_r=0.004 \text{ at 15 atm,}$$

$$\pi_r=0.008 \text{ at 60 atm,}$$

Now the motor is planned to be used in a desert environment in Arizona, where the ambient temperature reaches up to $T=45^\circ\text{C}$. The engineers have reported stable cases in this condition.

- a) Prove this mathematically.
- b) Find the $T_{\max}$ for this rocket to work fine.

2- Design Problem of a 2-Stage Rocket

Consider a design problem of a 2-stage rocket. The rocket is similarly staged with:

- $\varepsilon=\lambda=0.12$
- $M_L=50,000 \text{ Kg}$

The propellant composition is as follows:

- Fuel: 40%
 - 25% Aluminum Al (cr)
 - 15% HTPB (C_4H_6 butadiene)
- Oxidizer: 60%
 - 50% NH_3
 - 50% HNO_3

Operating Conditions:

- $P_c=85 \text{ atm}$

- $A_e/A^*=18$
- $\rho_p=1800 \text{ kg/m}^3$
- Propellant temperature: $T=298.15 \text{ K}$,

(a) Run CEA and find the following:

- Combustion temperature T_0
- Characteristic velocity c^*
- Thrust coefficient C_F
- Ratio of specific heats γ
- Specific heat C_p
- Mole fractions of main species

(b) Compute the following:

- Δu_b
- M_{01}, M_{02}
- M_{s1}, M_{s1} — structural mass
- M_{p1}, M_{p2} — propellant mass

(c) Find the propellant volume for both stages.

(d) Determine the throat area A^* for the first stage such that the thrust $T=3.1W$.

(e) Assume the propellant for each stage is a solid cylinder with the same diameter as the first stage nozzle exit. Find the height of the propellant in each stage.